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Reducing semantic ambiguity in open-vocabulary remote sensing image segmentation via knowledge graph-enhanced class representations

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ABSTRACT

Open-vocabulary semantic segmentation (OVSS) task presents a new challenge for remote sensing image understanding by requiring the recognition of previously unseen or novel classes during inference. However, existing OVSS methods often suffer from severe semantic ambiguity in land cover classification due to inconsistent naming conventions, hierarchical dependency, and insufficient semantic proximity in the embedding space. To address these issues, we propose KG-OVRSeg, a novel framework that mitigates semantic ambiguity by aggregating structured knowledge from a knowledge graph. This approach significantly enhances intra-class compactness and inter-class separability in the embedding space, thereby fundamentally enhancing class representations. We design a knowledge graph-enhanced class encoder (KGCE) that generates enriched class embeddings by querying hypernym–hyponym and synonym relationships within a localized knowledge graph. These enhanced embeddings are further utilized by a class attention gradual decoder (CAGD), which leverages a class-aware attention mechanism and guidance refinement to guide feature decoding. Extensive experiments on seven publicly available datasets demonstrated that KG-OVRSeg achieves state-of-the-art performance, with a mean mF1 of 51.65% and a mean mIoU of 39.18%, surpassing previous methods by 8.06% mF1 and 6.52% mIoU. Comprehensive ablation and visual analyses confirmed that KGCE significantly improves intra-class semantic compactness and inter-class separability in the embedding space, playing a crucial role in mitigating semantic inconsistency. Our work offers a robust and scalable solution for ambiguity-aware open-vocabulary tasks in remote sensing. The code is publicly available at https://github.com/HuangWBill/KG-OVRSeg.

1. Introduction

Remote sensing image semantic segmentation (RSISS) is a fundamental task in the remote sensing field, aiming to assign a semantic label to each pixel in an aerial or satellite image (Huang et al., 2023; Huang et al., 2024). RSISS has been widely applied in urban planning (Wei et al., 2020), land cover classification (Zermatten et al., 2025), and disaster assessment (Yang et al., 2024). With the rapid development of deep learning, RSISS methods have evolved from convolutional neural networks (Kampffmeyer et al., 2016; Wang et al., 2024a), to self-attention-based Transformer architectures (Wang et al., 2022; Zhang et al., 2022), and to the state-space modeling Mamba architecture (Ma et al., 2024). These advancements have driven RSISS towards higher accuracy and greater intelligence. Among these, the supervised deep learning methods based on the encoder-decoder architecture are the most common and widely studied. Due to the inherent characteristics of remote sensing images, challenges such as annotation difficulty, boundary ambiguity, and the limited performance of single modal approaches have increasingly emerged. These innovative solutions have been proposed, including semi-supervised learning (Zhang et al., 2023), edge-aware optimization (Chen et al., 2024; Huang et al., 2025), and multimodal fusion strategies (Wang et al., 2023a; Yang et al., 2021).

However, the aforementioned methods are constrained to closed-set segmentation, where models are trained and tested on a fixed set of predefined classes. This closed-world assumption hinders the scalability and flexibility of models in real-world deployments, where class definitions vary across datasets and unseen classes frequently occur (Zhu and Chen, 2024). To address this, inspired by large-scale vision-language pretraining models such as Contrastive Language-Image Pre-training model (CLIP, (Radford et al., 2021), researchers have introduced the open-vocabulary semantic segmentation (OVSS) task. Unlike traditional closed-set segmentation, OVSS enables models to generalize

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to unseen classes based on semantic relationships between class words, and without requiring additional annotations. In this setting, class words are embedded by a text encoder and aligned with image features in a shared semantic space to support pixel-level predictions for any classes (Wu et al., 2024).

Despite the success of OVSS in natural images, direct application to remote sensing images faces multiple unique challenges, primarily due to the inherent ambiguity and semantic complexity of remote sensing classes. Unlike objects that are clearly defined and have typical visual forms in natural images, land cover classes in remote sensing images often exhibit semantic ambiguity and representation uncertainty, which can be summarized in three main aspects:

(1) One-to-Many Mapping: Some high-level semantic classes contain multiple sub-classes, making it difficult for a single label word to capture their full visual characteristics. For example, the impervious surface class may refer to roads, parking lots, or roofs, while open space class could include green spaces, roads, or water bodies.
(2) Many-to-One Mapping: Different datasets or annotators may use various labels to refer to the same land cover class, introducing ambiguity. For instance, rangeland and pasture, barren and bareland, or cropland and plowed land may describe the same class but differ significantly in their text embeddings.
(3) Weak Semantic Proximity: Semantically similar classes may be far apart in the embedding space, distorting their actual semantic structure and damaging the generalization of model to unseen classes.

These challenges result in increased uncertainty and semantic ambiguity. As shown in Fig. 1, the state-of-the-art OVSS model (SegEarth-OV (Li et al., 2025b)) often difficult to accurately segment classes when only single label word is used. Improved performance is observed when using sub-class names or semantically related words, indirectly reflecting the limitations of current class representations. Fig. 2 illustrates those current methods (Ghiasi et al., 2022; Liang et al., 2023; Xu et al., 2023) typically rely on CLIP's text encoder to generate embeddings for single class words, ignoring the inherent hierarchical structure and semantic relationships within land cover classes. This leads to semantic misalignment and confusion, especially for unseen or ambiguous classes.

To address the aforementioned challenges, we propose a novel framework named Knowledge Graph-enhanced Open-Vocabulary Remote Sensing Semantic Segmentation (KG-OVRSeg). We introduce a hierarchical Land Cover Concept Knowledge Graph (LCCKG) to facilitate more structured and discriminative class embeddings. By constructing subgraphs that capture both hypernym–hyponym and synonym relationships, our method enhances the structural consistency and semantic clarity of class representations, thereby improving the model's ability to recognize ambiguous classes. These enriched embeddings are then fed into a class attention gradual decoder, where the class-aware attention (CAA) module and the guidance refine (GR) module impose semantic constraints throughout the decoding process. This design effectively reduces semantic confusion and improves prediction accuracy for both seen and unseen classes. Furthermore, to suppress background interference during inference, we introduce a background threshold strategy that enables more effective distinction between foreground and background classes. The main contributions of this paper are summarized as follows:

(1) We propose a knowledge graph-enhanced class encoder (KGCE) that constructs a knowledge subgraph for each class word by using hierarchical and synonym relationships from the LCCKG. Through graph-based embedding aggregation, KGCE produces more discriminative class representations, guiding the model to generate more accurate class-specific responses in the image feature space and effectively mitigating embedding inconsistencies and semantic ambiguity.
(2) We design a class attention gradual decoder (CAGD), which uses enriched class embeddings to guide the multi-stage decoding process via a class-aware attention mechanism. This enhances spatial response to target-relevant regions, achieving more accurate and fine-grained segmentations.
(3) We conduct extensive experiments on multiple RSISS datasets for OVSS, demonstrating that KG-OVRSeg significantly outperforms state-of-the-art methods, particularly under challenging unseen-class scenarios. Furthermore, the proposed KGCE can be freely integrated into existing remote sensing OVSS pipelines, substantially improving their open-vocabulary capabilities.

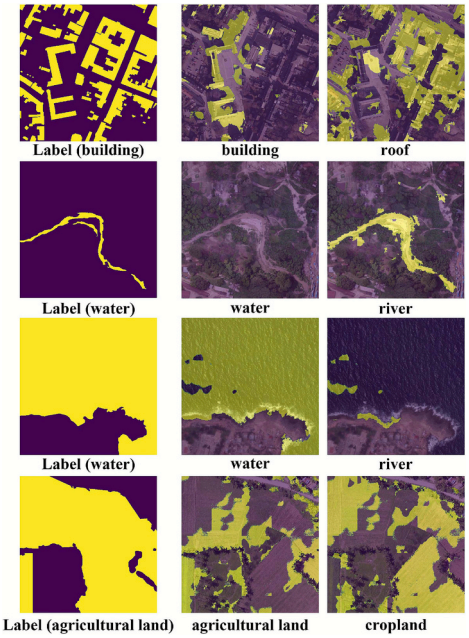

Fig. 1. Text-image response relationship of class words in SegEarth-OV. The first column represents the true labeling results, while the second and third columns represent the segmentation results obtained using different class words.

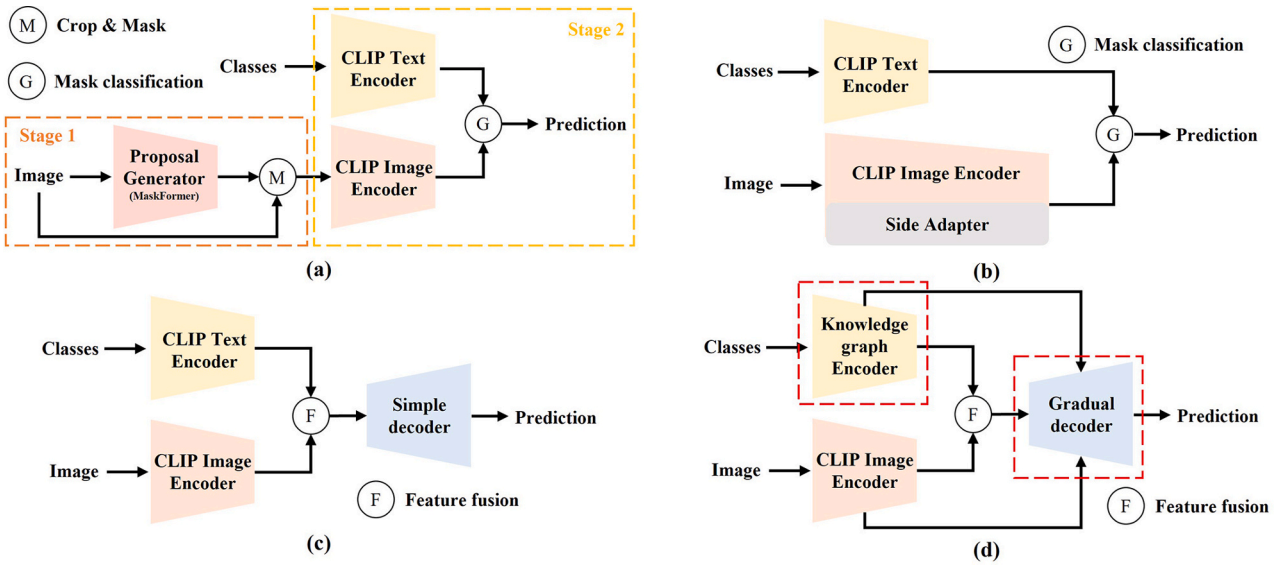

Fig. 2. Comparison of the proposed framework with the previous frameworks: (a) Two-stage method, (b) side adapter method, (c) single-stage method, (d) our proposed KG-OVRSeg method.

The remainder of this paper is organized as follows: Section 2 reviews related work on OVSS and semantic ambiguity handling in remote sensing. Section 3 details the proposed KG-OVRSeg framework. Section 4 introduces the LCCKG, datasets, experimental settings, and evaluation metrics. Section 5 presents experimental results and analysis. Section 6 discusses the effectiveness and response relationship of knowledge-guided embeddings. Finally, Section 7 concludes the paper.

2. Related work

2.1. Open-vocabulary semantic segmentation

OVSS has emerged as a promising research direction, enabling models to segment any classes in any image without manual annotations. This capability is largely attributed to the advent of vision-language pretrained models such as CLIP (Radford et al., 2021) and BLIP (Li et al., 2022b), which jointly embed images and text into a shared feature space.

As shown in Fig. 2, early OVSS frameworks adopt a two-stage design. In the first stage, a mask proposal model (e.g., MaskFormer (Cheng et al., 2021)) is trained to generate class-agnostic mask proposals. In the second stage, each proposal is matched to the most relevant class embedding generated by CLIP's text encoder. Representative works include ZSSeg (Xu et al., 2022), OpenSeg (Ghiasi et al., 2022), and OVSeg (Liang et al., 2023). DeOP (Han et al., 2023) replaces CLIP's self-attention layers with mask proposals to generate anchor heatmaps, which are then matched with class embeddings. Despite their effectiveness, these two-stage methods tend to be architecturally complex and computationally expensive.

To address these limitations, lightweight adapter-based methods have been proposed. SAN (Xu et al., 2023) models semantic segmentation tasks as a region recognition problem by adding side adapters. The two branches of the side adapter are used to predict mask proposals and attention bias, respectively. SemLA (Qorbani et al., 2025) stores LoRA

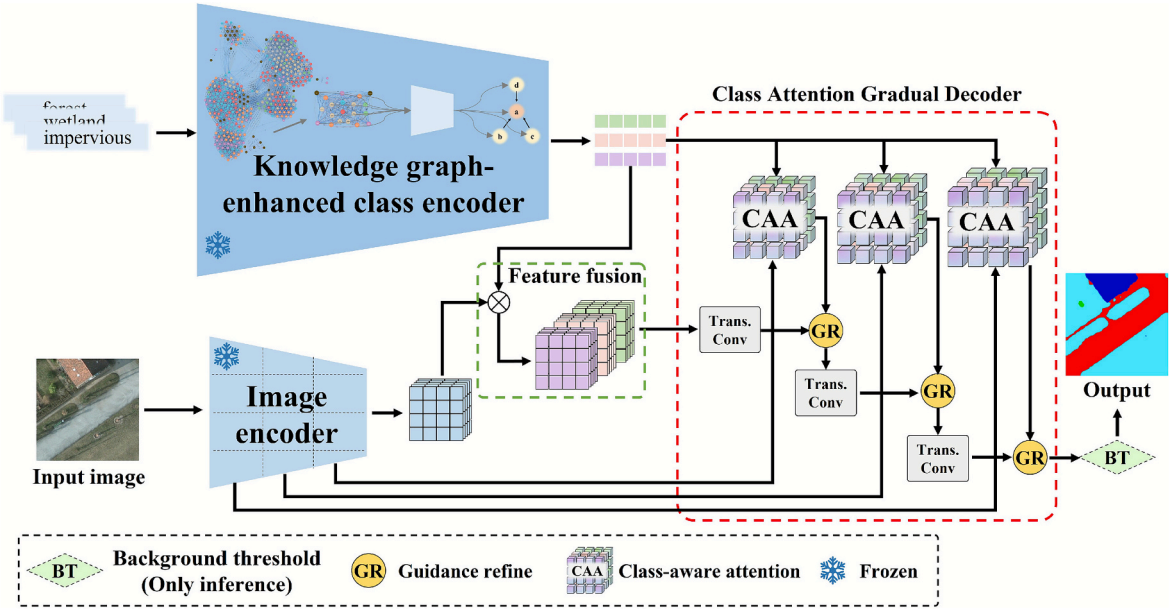

Fig. 3. Overall architecture of the proposed KG-OVRSeg method.

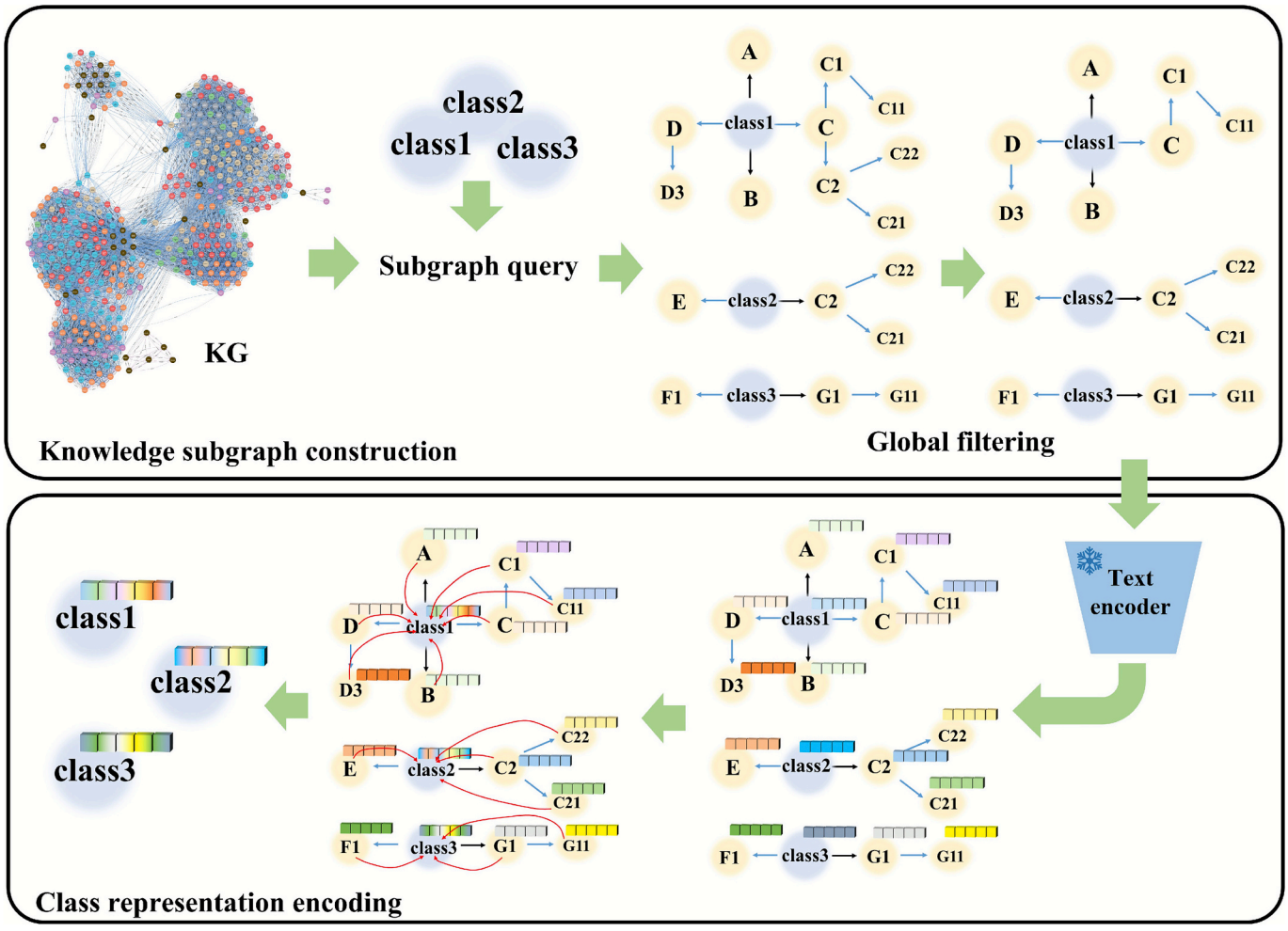

Fig. 4. The pipeline of knowledge graph-enhanced class encoder.

adapters for per domain, enabling modular inference without retraining.

Single-stage OVSS frameworks offer greater efficiency. LSeg (Li et al., 2022a) is the first pixel-level single-stage model. It encodes both visual and text features, aligns them via feature fusion, and applies spatial regularization to enhance semantic consistency around object boundaries. Subsequent efforts focus on improving multi-modal fusion modules. Fusioner (Ma et al., 2022) introduces a lightweight cross-modal transformer fusion module that aligns frozen visual features with textual embeddings. Recently, Cat-Seg (Cho et al., 2024) pushes single-stage frameworks to a new level. It proposes a new cost aggregation strategy to guide inference and decoding. Based on the framework, SED (Xie et al., 2024) proposes early class rejection and skip-layer fusion module to refine feature fusion further. FGAseg (Li et al., 2025a) refines CLIP-based alignment through cross-modal attention and text-pixel alignment losses, also proposing a category information supplementation module to obtain detailed boundary information. These methods have greatly improved OVSS performance on natural images and laid the foundation for exploring OVSS in remote sensing.

2.2. Open-vocabulary semantic segmentation for remote sensing

Research on OVSS in remote sensing is still in its early stages of development. The earliest work (Chen and Bruzzone, 2023) employed a U-Net to produce segmentation masks, then aligned the encoded features with CLIP embeddings via contrastive loss. This marked the first application of OVSS to remote sensing. Subsequent studies have extended general-domain frameworks like Cat-Seg to better suit remote sensing-specific challenges. For instance, OVRS (Cao et al., 2025) introduced a rotation-aggregative similarity computation module and a multi-scale feature integration strategy to accommodate variations in object orientation and scale. GSNet (Ye et al., 2025) added self-supervised DINO features as domain experts to guide general feature fusion. However, these supervised models require training with existing remote sensing datasets to learn the feature fusion and decoding modules, which limits their scalability in label-scarce scenarios. To mitigate this dependency, SegEarth-OV (Li et al., 2025b) was proposed, which is a training-free method that learns a simple and universal upsampler (SimFeatUp) through self-supervised feature reconstruction. This method is easy to use and maintains average performance on various datasets, but its performance is limited and lacks flexibility for retraining and fine-tuning. Crucially, all existing methods directly apply the model to the remote sensing field using remote sensing image data during training, overlook the semantic ambiguity among land cover classes, and fail to address class-level inconsistencies inherent in remote sensing data.

2.3. Class representation and semantic ambiguity handling in remote sensing

To handle the unique characteristics of remote sensing imagery, some researchers have attempted to fine-tune CLIP models using remote sensing image-text pair datasets, resulting in domain-specific vision-language models (VLMs). These VLMs have been applied to tasks such as zero-shot image classification, text retrieval, and object counting.

RemoteCLIP (Liu et al., 2024) is the first remote sensing VLM. It organizes and constructs large-scale image-text pairs based on existing remote sensing datasets to fine-tune the CLIP model. Subsequently, GeoRSCLIP (Zhang et al., 2024) fine-tuned the CLIP model on the constructed RS5M dataset. Similarly, SkyCLIP (Wang et al., 2024b) used Google Earth Engine (GEE) images and OpenStreetMap semantic information to build the SkyScript dataset for training. While these remote sensing VLMs have demonstrated effectiveness in non-dense prediction tasks such as classification, their application to dense prediction tasks like segmentation remains challenging due to spatial and semantic complexity.

In RSISS, Li et al. (2022c) used an ontology-based knowledge reasoning tool to perform post-processing optimization based on predefined ontology rules. Although this approach can alleviate semantic ambiguity to some extent, it lacks scalability to unseen classes and requires extensive manual ontology design. Recently, TACOSS (Zermatten et al., 2025) conducted a detailed study on class label representation methods in remote sensing OVSS tasks. It analyzed the impact of various text descriptors (e.g., class names, synonyms, sentence descriptions) and text embedding methods on OVSS performance. Although it offers insightful guidance for label embedding in remote sensing, it still depends on manually extending class descriptions, which is time-consuming and inherently uncertain.

2.4. Knowledge graph and its applications

Knowledge Graphs (KGs) is a structured semantic network that represents real-world entities and their interrelationships in the form of triples (Ji et al., 2022). It provide an effective method of knowledge representation. Since Google formally introduced the concept of the Knowledge Graph in 2012, more and more research has focused on KG construction, knowledge representation, and various downstream applications (Hao et al., 2021).

In general, KGs are mainly divided into general KGs and domain KGs. In the early stages of general KGs construction, entities and relations were manually extracted. With the development of natural language processing (NLP) technology, researchers began using deep learning models trained on large-scale text corpora to automatically extract entities, relations, and attributes (Devlin et al., 2019; Yao et al., 2019; Zhu et al., 2021). Recently, the emergence of large language models (LLMs) has further accelerated this process, enabling automated KG construction (Sun et al., 2025; Yao et al., 2025) and updating (Yang et al., 2025). Beyond the aforementioned studies, KGs have also been applied to vision-related tasks, such as scene graph generation (Chen et al., 2019), visual question answering (Ding et al., 2022), and image classification (Marino et al., 2017), where structured prior knowledge helps enhance visual reasoning and semantic understanding.

In the field of remote sensing, Li et al. (2021) constructed the first Remote Sensing Knowledge Graph (RSKG) and applied it to zero-shot classification in RS scenes. Although this initial KG was relatively small in scale, it demonstrated the feasibility of incorporating structured knowledge into remote sensing task. Later, Li et al. (2024) developed STAR, the first large-scale scene graph generation dataset for satellite imagery, comprising over 210,000 objects and 400,000 triples. Li et al. (2025c) proposed a multi-granularity knowledge graph reasoning (mKGR) framework that automatically establishes relationships between multimodal geospatial datasets and land-use classes, enabling zero-shot land-use mapping. Additionally, several domain KGs have been constructed and successfully applied to specialized RS tasks, including urban flood risk assessment (Wang et al., 2023b), mountain vegetation classification (Yao and Liu, 2024), and hyperspectral image classification (Lei et al., 2025). These studies have demonstrated that the incorporation of prior knowledge from KGs can significantly enhance model performance. However, most existing KG-driven approaches are limited to image-level or scene-level tasks, while their applications at the pixel-level remain in their infancy.

3. Methodology

This section describes the details of the proposed KG-OVRSeg framework, shown in Fig. 3. Given a remote sensing image and a set of open-vocabulary class names, KG-OVRSeg predicts pixel-wise semantic labels for both seen and unseen categories. To achieve this, we leverage two frozen pre-trained encoders: one for the image and one for the text. For each class name, we use it to query a structured land cover concept knowledge graph and retrieve a semantically relevant subgraph. The subgraph is encoded into a knowledge-aware class embedding, which is then used to guide the decoder via attention mechanisms at multiple stages.

KG-OVRSeg consists of four main components: (1) an image encoder detailed in Section 3.1. (2) a knowledge graph-enhanced class encoder, as detailed in Section 3.2. (3) a feature fusion module defined in Section 3.3, and (4) a class attention gradual decoder described in Section 3.4. Finally, Section 3.5 provides a detailed description of the open-vocabulary inference process of the model.

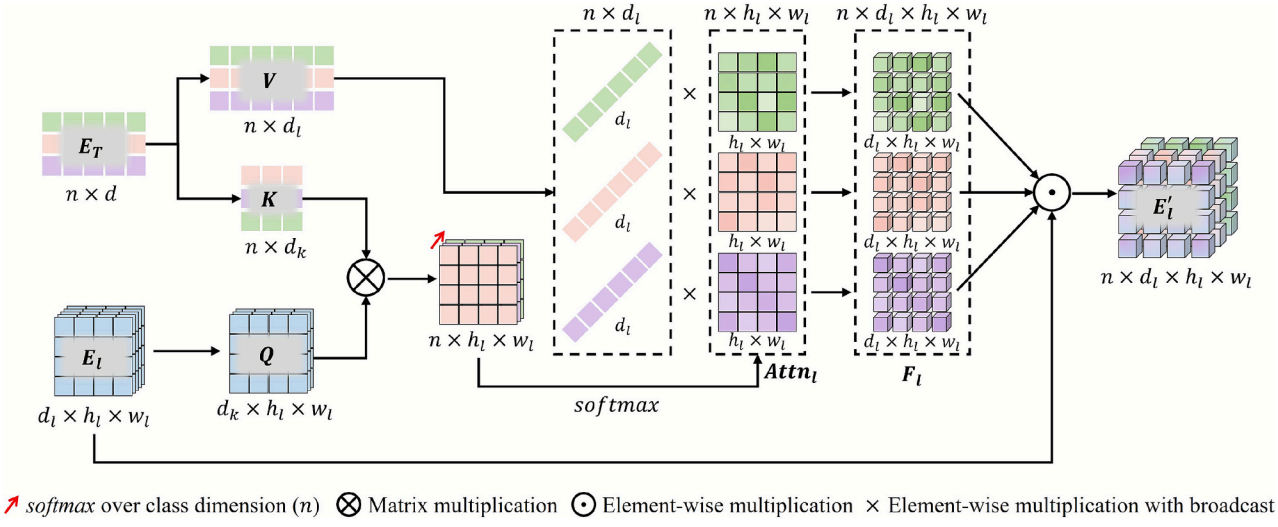

Fig. 5. Architecture of the class-aware attention module.

3.1. Image encoder

The image encoder employs a visual backbone to extract multi-level visual features from the input image. Following the standard visual encoder architecture adopted in current vision-language pre-trained models, we utilize the ViT-B/16 architecture as our backbone. Let $I \in \mathbb{R}^{H\times W\times 3}$ (with H = height and W = width) be the input image. The encoder outputs a feature map $E_I \in \mathbb{R}^{d\times h\times w}$, where d, h and w represent the feature dimension, height and width, respectively. To support effective decoding, multi-level features are extracted from intermediate transformer blocks. These features are passed into the decoder alongside class-aware signals to facilitate spatial-semantic alignment.

3.2. Knowledge graph-enhanced class encoder

As shown in Fig. 4, the KGCE module uses structured information from a knowledge graph to enhance the semantic representations of class labels. This mechanism helps mitigate semantic ambiguity and poor generalization. The KGCE process includes two main stages: knowledge subgraph construction, and class representation encoding.

3.2.1. Knowledge subgraph construction

Given any set of target class labels $C = \{c_1, c_2, \cdots, c_n\}$ for segmentation, we first query a pre-constructed knowledge graph to extract a semantically relevant local subgraph for each class word. These subgraphs capture the contextual knowledge associated with each class in a structured form, serving as the foundation for further representation enhancement.

Specifically, for each class term c_i, we perform an exact match in the KG to locate the corresponding node, then extract its related subgraph based on predefined relation types r. Formally, for class c_i, the retrieved subgraph is denoted as:

$$\mathscr{G}_i = (\mathscr{V}_i, \mathscr{R}, \mathscr{T}_i) \tag{1}$$

where $\mathscr{V}_i$ denotes the set of all entity nodes included in the subgraph, and $\mathscr{T}_i = \{(h, r, t) | h \in \mathscr{V}_i, r \in \mathscr{R}, t \in \mathscr{V}_i\}$ represents the collection of all knowledge triples. Each triple (h, r, t) consists of a head entity h, a relation r, and a tail entity t, together forming the semantic context of the class label within the knowledge graph.

In the class set C, semantics contain relationships that may exist among class words. For instance, when the target classes include "agricultural land", "vineyards", and "plowed land", it is crucial to recognize that "agricultural land" is a parent class encompassing the other two. In such cases, directly extracting and retaining the subgraphs for all three classes can result in semantic redundancy, where the subgraph of a higher-level class ("agricultural land") implicitly includes information from its subclasses ("vineyards" and "plowed land"). This redundancy can blur the semantic boundaries of general classes and impair the model's ability to distinguish between them.

To eliminate redundant knowledge across overlapping classes, we propose a global filtering strategy that filters out subgraph structures embedded within higher-level classes. Given the set of initially constructed class-specific subgraphs $\mathscr{G} = \{\mathscr{G}_1, \mathscr{G}_2, \cdots, \mathscr{G}_n\}$, we perform pairwise checks among all classes to detect contain relationships. For any two classes c_i, $c_j \in C$, if c_i semantically contains c_j, and the corresponding subgraphs satisfy:

$$\mathscr{V}_j \subseteq \mathscr{V}_i, \mathscr{T}_j \subseteq \mathscr{T}_i \tag{2}$$

then subgraph $\mathscr{G}_j$ is considered to be completely embedded within subgraph $\mathscr{G}_i$.

To avoid semantic overlap, we subtract the elements of $\mathscr{G}_j$ from $\mathscr{G}_i$, resulting in a refined subgraph:

$$\mathscr{G}_i' = \left(\mathscr{V}_i - \mathscr{V}_j, \mathscr{R}, \mathscr{T}_i - \mathscr{T}_j\right) \tag{3}$$

After applying this filtering process to all relevant class pairs, we obtain an updated subgraph set $\mathscr{G}' = \{\mathscr{G}_1', \mathscr{G}_2', \cdots, \mathscr{G}_n'\}$. This ensures that each class is associated with a semantically unique and discriminative subgraph, avoiding ambiguity and improving the robustness of downstream representation learning.

3.2.2. Class representation encoding

After determining the final filtered subgraph for each class, we perform semantic encoding on all constituent nodes to obtain enriched class-level representations. This step is crucial for injecting structured knowledge into the visual-language alignment process.

Specifically, for each refined class-specific subgraph $\mathscr{G}_i' = (\mathscr{V}_i', \mathscr{R}, \mathscr{T}_i')$, we employ a pre-trained CLIP text encoder to generate text embeddings for each node $v_i' \in \mathscr{V}_i'$, The embedding for node v_i' is computed as:

$$h_i = CLIPTextEncoder(v_i') \in \mathbb{R}^d \tag{4}$$

where d is the embedding dimension. All node embeddings within the subgraph are then stacked to form an initial subgraph representation matrix:

$$\mathscr{H}_i = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_{\mathscr{V}_i'} \end{bmatrix} \in \mathbb{R}^{|\mathscr{V}_i'|\times d} \tag{5}$$

We apply average pooling to summarize all node features into a single vector representing the entire subgraph:

$$E_T^i = Avgpool(\mathscr{H}_i) \in \mathbb{R}^d \tag{6}$$

Finally, $E_T \in \mathbb{R}^{n\times d}$ is the knowledge-enhanced semantic embedding, the enriched class embedding is later fused with the spatial features from the image decoder, enabling the model to perform robust segmentation even on complex or previously unseen classes.

3.3. Feature fusion

After obtaining the class embeddings $E_T \in \mathbb{R}^{n\times d}$ from the KGCE module and the image features $E_I \in \mathbb{R}^{d\times h\times w}$ from the visual encoder, we aim to align and fuse these two modalities at the spatial level for fine-grained semantic understanding. To enable spatial alignment, we employ a spatial broadcasting mechanism that expands each class embedding E_T to match the spatial dimensions of the image features, resulting in $E_T' \in \mathbb{R}^{n\times d\times 1\times 1}$. Subsequently, a dot-product operation is performed between the image feature map E_I and the broadcasted class embeddings E_T', enabling semantic interaction at each spatial location.

$$S = E_T' \odot E_I \tag{7}$$

where $\odot$ represents element-wise multiplication. $S \in \mathbb{R}^{n\times d\times h\times w}$ represents the class-to-pixel similarity map, indicating how strongly each class embedding correlates with the visual representation at every location.

3.4. Class attention gradual decoder

We propose a CAGD to progressively guide the decoder towards high-resolution semantic reconstruction. This module takes as input the similarity map $S \in \mathbb{R}^{n\times d\times h\times w}$, multi-level image features $E_l \in \mathbb{R}^{d_l\times h_l\times w_l}$, and the enhanced class embeddings $E_T \in \mathbb{R}^{n\times d}$. It first applies a CAA mechanism to enhance semantic representations by reinforcing spatial regions relevant to each target class. Subsequently, a GR module is employed to progressively recover spatial structure, using the class attention maps to refine the similarity representation.

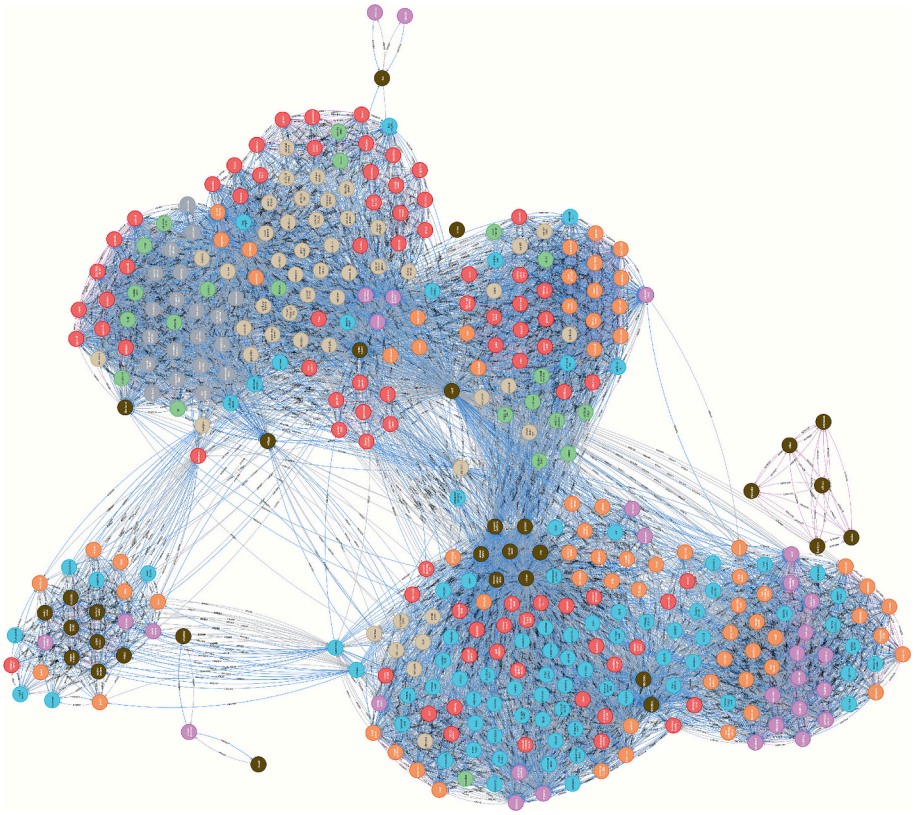

Fig. 6. Illustration of the Land Cover Concept Knowledge Graph (LCCKG). The circles of different colors represent nodes, and the lines of different colors between nodes represent relationships. The node consists of 383 land cover classes, with four types of relationships: synonym, contain, subclass, and different classes.

3.4.1. Class-aware attention

The CAA module is designed to enhance spatial features based on class-specific regions. As shown in Fig. 5, the CAA module takes the enhanced class embeddings $E_T \in \mathbb{R}^{n\times d}$ and the multi-level image features $E_l \in \mathbb{R}^{d_l\times h_l\times w_l}$ as input. First, E_T is projected into a key matrix $K \in \mathbb{R}^{n\times d_k}$ and a value matrix $V \in \mathbb{R}^{n\times d_l}$ via linear transformations:

$$K = E_T W^K \tag{8}$$

$$V = E_T W^V \tag{9}$$

where W^K and W^V are learnable weight matrices.

To reduce computation cost, the E_l is reshaped from (d_l, h_l, w_l) to $(h_l w_l,\ d_l)$ and then projected into the query space through another learnable matrix W^Q:

$$Q = reshape(E_l)W^Q \tag{10}$$

followed by reshaping Q back to (d_k, h_l, w_l).

The attention map $Attn_l \in \mathbb{R}^{n\times h_l\times w_l}$ is calculated using matrix multiplication ($\otimes$) between K and Q, and applying a *softmax* operation over the class dimension n:

$$Attn_l = softmax(K \otimes Q, overclassdimension(n)) \tag{11}$$

where each attention score represents the relevance between a spatial position and a class embedding.

The attention weights are used to aggregate the value matrix V for each spatial location along the class dimension, resulting in class-aware spatial features $F_l \in \mathbb{R}^{n\times d_l\times h_l\times w_l}$:

$$F_l = V \times Attn_l \tag{12}$$

where $\times$ represents element-wise multiplication with broadcast.

Finally, F_l are fused with the original spatial features E_l through element-wise multiplication ($\odot$):

$$E_l' = F_l \odot E_l \tag{13}$$

The output $E_l' \in \mathbb{R}^{n\times d_l\times h_l\times w_l}$ is class-enhanced spatial representations. This operation enhances the model's response to class-specific regions guided by knowledge-derived class priors, thereby improving fine-grained segmentation performance.

3.4.2. Guidance refine

To effectively fuse high-level semantics with low-level spatial details during decoding, we introduce a GR module that performs guided upsampling and structural refinement. Before applying GR, the $S \in \mathbb{R}^{n\times d\times h\times w}$ is first upsampled and channel-compressed via a transposed convolution to obtain $S_U \in \mathbb{R}^{n\times(d-d_l)\times h_l\times w_l}$:

$$S_U = ConvTranspose2d(S) \tag{14}$$

The output feature maps S_U shares the same spatial resolution as E_l'. The two features are concatenated along the channel dimension to obtain $S_{Ul}' \in \mathbb{R}^{n\times d\times h_l\times w_l}$:

$$S_{Ul}' = Concat(S_U, E_l') \tag{15}$$

The concatenated feature is passed through two consecutive 3×3 convolution layers with group normalization, reducing channel dimensions and refining feature representation:

$$S_l = Conv2d(Conv2d(S_{Ul}')) \tag{16}$$

The $S_l \in \mathbb{R}^{n\times d_l\times h_l\times w_l}$ is used as the updated S for the next decoding stage. This module plays a critical role in progressively restoring spatial structure while integrating semantic guidance.

Table 1
Comparison of some basic information in different datasets.

Dataset	Train	Inference	Spatial resolution	Classes
OpenEarthMap (train)	✓		0.25–0.5 m	9 classes: bareland, rangeland, developed space, road, tree, water, agriculture land, building, background
OpenEarthMap (test)		✓	0.25–0.5 m	9 classes: bareland, rangeland, developed space, road, tree, water, agriculture land, building, background
Potsdam (test)		✓	0.05 m	6 classes: impervious surfaces, building, low vegetation, tree, car, background
LoveDA (test)		✓	0.3 m	7 classes: buildings, road, water, barren, forest, agriculture, background
GID Large (test)		✓	0.8 m	6 classes: built up, farmland, forest, meadow, water, background
Globe230k (test)		✓	1 m	11 classes: cropland, forest, grass, shrubland, wetland, water, tundra, impervious, bareland, ice, background
FLAIR #1 (test)		✓	0.2 m	13 classes: building, pervious surface, impervious surface, bare soil, water, coniferous, deciduous, brushwood, vineyard, herbaceous vegetation, agricultural land, plowed land, background
LandCover.ai (test)		✓	0.25–0.5 m	5 classes: buildings, woodlands, water, roads, background

Table 2
Quantitative comparison of different methods. The best and second-best results are highlighted in **bold** and underlined, respectively.

Method	*mF*1/*mIoU*(%)							*meanmF*1(%)	*meanmIoU*(%)
	Potsdam	LoveDA	GID Large	Globe230k	FLAIR #1	OpenEarthMap	LandCover.ai		
LSeg	45.87/34.79	46.74/32.75	34.88/23.53	15.70/10.04	24.89/16.13	**79.08/66.66**	34.81/26.24	40.28	30.02
Fusioner	39.68/28.53	48.95/35.00	38.01/27.72	19.18/12.59	19.50/13.62	70.22/55.87	38.08/25.61	39.09	28.42
SAN	34.65/19.52	47.79/33.18	41.79/28.35	20.06/13.31	19.78/12.93	62.31/47.60	**60.01/44.61**	40.91	28.50
Cat-Seg	40.37/29.44	47.11/33.48	42.14/31.80	30.92/22.48	24.08/16.08	71.64/57.54	35.58/24.48	41.69	30.76
SegEarth-OV	46.43/33.46	38.84/26.45	40.88/29.81	28.72/19.74	23.48/15.60	44.50/32.07	41.68/30.84	37.79	26.85
OVRS	37.28/27.16	47.62/34.13	42.54/32.42	29.96/21.58	26.97/18.12	71.27/57.25	39.43/28.21	42.15	31.27
FGA-Seg	42.27/30.94	51.10/37.05	42.32/31.42	29.41/20.22	**27.29/19.93**	71.06/57.16	41.86/31.87	43.59	32.66
TACOSS	34.73/24.58	41.33/29.60	26.31/17.13	16.84/11.19	19.01/13.90	76.90/64.14	45.08/32.26	37.17	27.40
Ours	**53.74/40.03**	**52.34/37.74**	**62.51/46.93**	**34.08/23.31**	26.30/19.61	77.33/64.59	55.27/42.03	**51.65**	**39.18**

3.5. Inference

Foreground and background separation is particularly challenging in remote sensing scenes due to blurred boundaries and mixed land cover types. During training, all unspecified classes are treated as background, which may lead to ambiguous background representations. However, in inference, some previously unseen background classes may correspond to actual foreground objects, and conversely, some training foreground classes may appear as background. This semantic mismatch between training and inference stages can result in false positives or false negatives.

During inference, the model receives an input image along with a list of target class words. To enable separation between foreground and background, an additional "background" class is appended to the class list. The trained model generates class-specific probability maps by fusing the visual features from the input image with the text representations of the provided target class words. Afterwards, to address the issue where the background class is mistakenly classified as the foreground class, we introduce a background thresholding (BT) module. This module is based on a fixed threshold parameter τ to measure the response strength of each spatial location across all predicted classes. If the maximum response of all classes at a certain pixel are below the τ, the pixel is classified as background.

4. Experimental settings

4.1. Knowledge graph

The knowledge graph used in this study is the Land Cover Concept Knowledge Graph (LCCKG), which we previously developed and publicly released [1]. As illustrated in Fig. 6, LCCKG consists of 383 land cover classes, 4 semantic relationship types, and 146,306 triplets. This graph was constructed through a comprehensive analysis and integration of published literature (from 2016 to 2024), publicly available datasets, and existing knowledge graphs. Domain experts manually annotated hierarchical relationships between any pairs of classes. The graph construction utilized 127 publicly available remote sensing semantic segmentation datasets, encompassing nearly all possible land cover classes. LCCKG also supports incremental updating, making it well-suited for adapting to previously unseen classes and expanding label coverage over time.

4.2. Datasets

We evaluate the proposed method on seven publicly available RSISS datasets, using one for training and the others for evaluating generalization performance. These datasets differ significantly in terms of source, spatial resolution, and number of semantic classes. Specifically, we use the training set of the OpenEarthMap dataset for model training, while the test sets of all seven datasets are used during inference. The resolution and class details for each dataset are summarized in Table 1. (1) OpenEarthMap Dataset (Xia et al., 2023) is a benchmark dataset for global high-resolution land cover mapping. It contains heterogeneous images from multiple platforms, covering 97 regions across 44 countries and 6 continents. (2) The Potsdam Dataset [2] is a typical historical city with large buildings, narrow streets, and dense settlements. It includes 38 remote sensing images, each sized 6000 × 6000 pixels. (3) LoveDA Dataset (Wang et al., 2021) is a domain-adaptive semantic segmentation dataset comprising 5987 high-resolution images from urban and rural areas of Nanjing, Guangzhou, and Wuhan. It features multi-scale objects, complex backgrounds, and class imbalance. (4) GID Large Dataset (Tong et al., 2020) is a large-scale land cover dataset constructed from Gaofen-2 satellite images, which has the advantages of large-scale coverage, wide distribution, and high-resolution. (5) Globe230k Dataset (Shi et al., 2023) is a massive-scale, globally distributed land cover dataset with

[1] https://github.com/HuangWBill/LCCKG_ERLG-update.

[2] https://www2.isprs.org/commissions/comm3/wg4/semantic-label-ing.html.

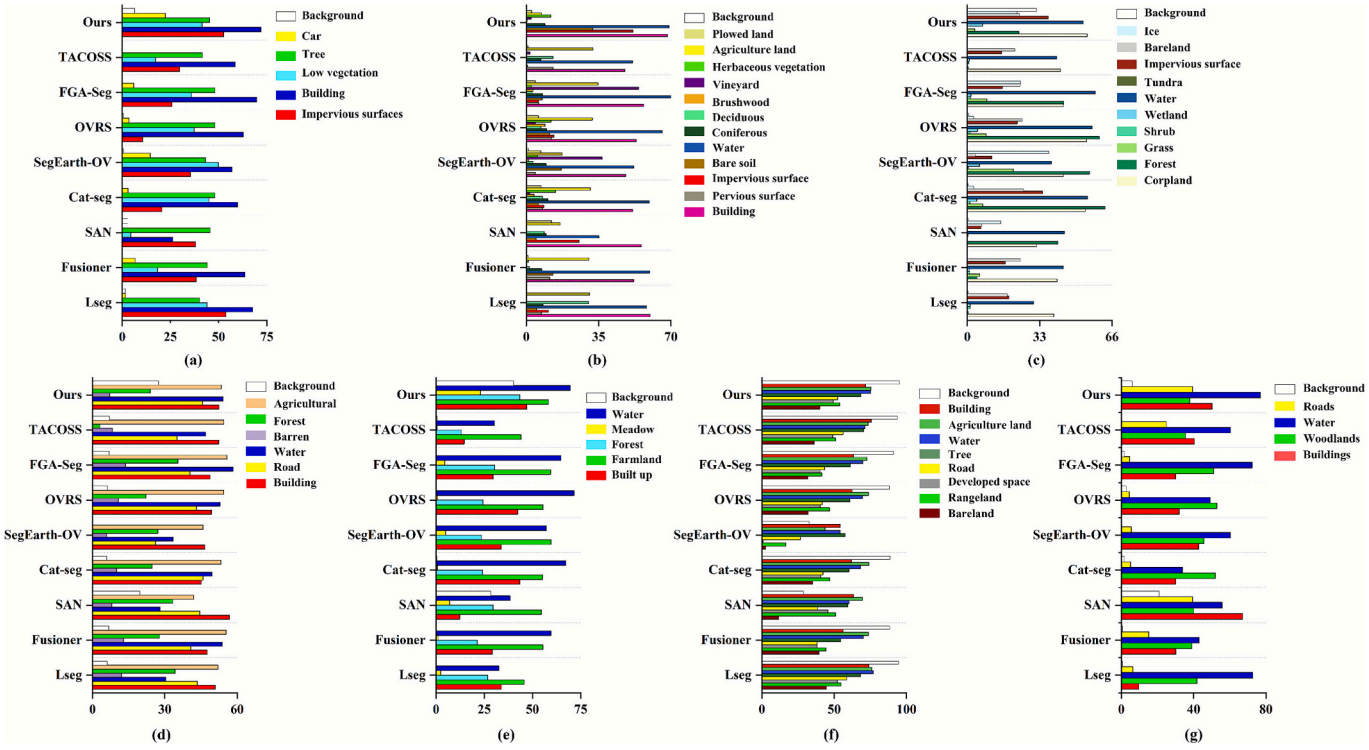

Fig. 7. Per-class IoU scores of different methods across datasets. (a) Potsdam, (b) FLAIR #1, (c) Globe230k, (d) LoveDA, (e) GID Large, (f) OpenEarthMap, (g) LandCover.ai.

high-resolution remote sensing images. It stands out for its large scale, semantic diversity, and multimodality. (6) FLAIR #1 Dataset (Garioud et al., 2022) is a part of the dataset currently used at IGN to establish the French national land cover map reference, covering 50 different spatial domains with high spatiotemporal heterogeneity. (7) LandCover.ai Dataset (Boguszewski et al., 2021) is an aerial image dataset covering 216.27 km^2 of rural Poland.

4.3. Implementation details

All experiments were conducted on a Linux PC with two Intel (R) Xeon (R) 6133 CPUs and four NVIDIA GeForce RTX 4090D GPUs, each with 24 GB of memory. During training, we apply the random scaling (scale factors from [0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0]), random cropping and random flipping for data augmentation. The model is optimized using *AdamW* with a poly decay learning rate schedule. The initial learning rate is set to 2e-5, with a weight decay of 0.0001, and a beta of (0.9, 0.999). The loss function selects cross-entropy loss. The image encoder is based on ViT-B/16, and we extract multi-level features from layers 2, 5, and 8 for gradual guidance. The corresponding feature dimensions d_l are 256, 128, and 64, respectively. The class embedding dimension d_k is set to 512. To fit GPU memory constraints, we set the batch size to 4 and train the model for a maximum of 80k iterations.

4.4. Evaluation metrics

We report performance using mean Intersection over Union (*mIoU*) and mean $F1$ score (*mF*1), the two most commonly adopted metrics in the RSISS task. These metrics provide a comprehensive evaluation from different perspectives. In addition to reporting *mF*1 and *mIoU* for each dataset individually, we also compute the *meanmF*1 and *meanmIoU* across all datasets to assess overall generalization performance. The specific definitions of the formulas for these metrics are as follows:

$$mF1 = \frac{1}{N}\sum_{k=1}^{N} \frac{2 \times Precision_k \times Recall_k}{Precision_k + Recall_k} \tag{15}$$

$$Precision_k = \frac{TP_k}{FP_k + TP_k} \tag{16}$$

$$Recall_k = \frac{TP_k}{FN_k + TP_k} \tag{17}$$

$$mIoU = \frac{1}{N}\sum_{k=1}^{N} IoU_k = \frac{1}{N}\sum_{k=1}^{N} \frac{TP_k}{TP_k + FP_k + FN_k} \tag{18}$$

$$meanmF1 = \frac{1}{M}\sum_{i=1}^{M} mF1_i \tag{19}$$

$$meanmIoU = \frac{1}{M}\sum_{i=1}^{M} mIoU_i \tag{20}$$

where TP_k, FP_k, TN_k, and FN_k denote true positives, false positives, true negatives, and false negatives for class k, respectively. N is the number of classes in a dataset, M represents the number of datasets.

5. Results

5.1. Comparison with state-of-the-art methods

To comprehensively evaluate the performance of KG-OVRSeg, we conducted a series of comparative experiments involving both classical OVSS models in computer vision (Lseg (Li et al., 2022a), Fusioner (Ma et al., 2022), SAN (Xu et al., 2023)) and the state-of-the-art methods (Cat-Seg (Cho et al., 2024), FGA-Seg (Li et al., 2025a)), as well as recent methods specifically designed for the remote sensing (SegEarth-OV (Li et al., 2025b), OVRS (Cao et al., 2025), TACOSS (Zermatten et al., 2025)).

Table 2 presents the quantitative evaluation results of different methods. It is evident that our method achieves the best results with

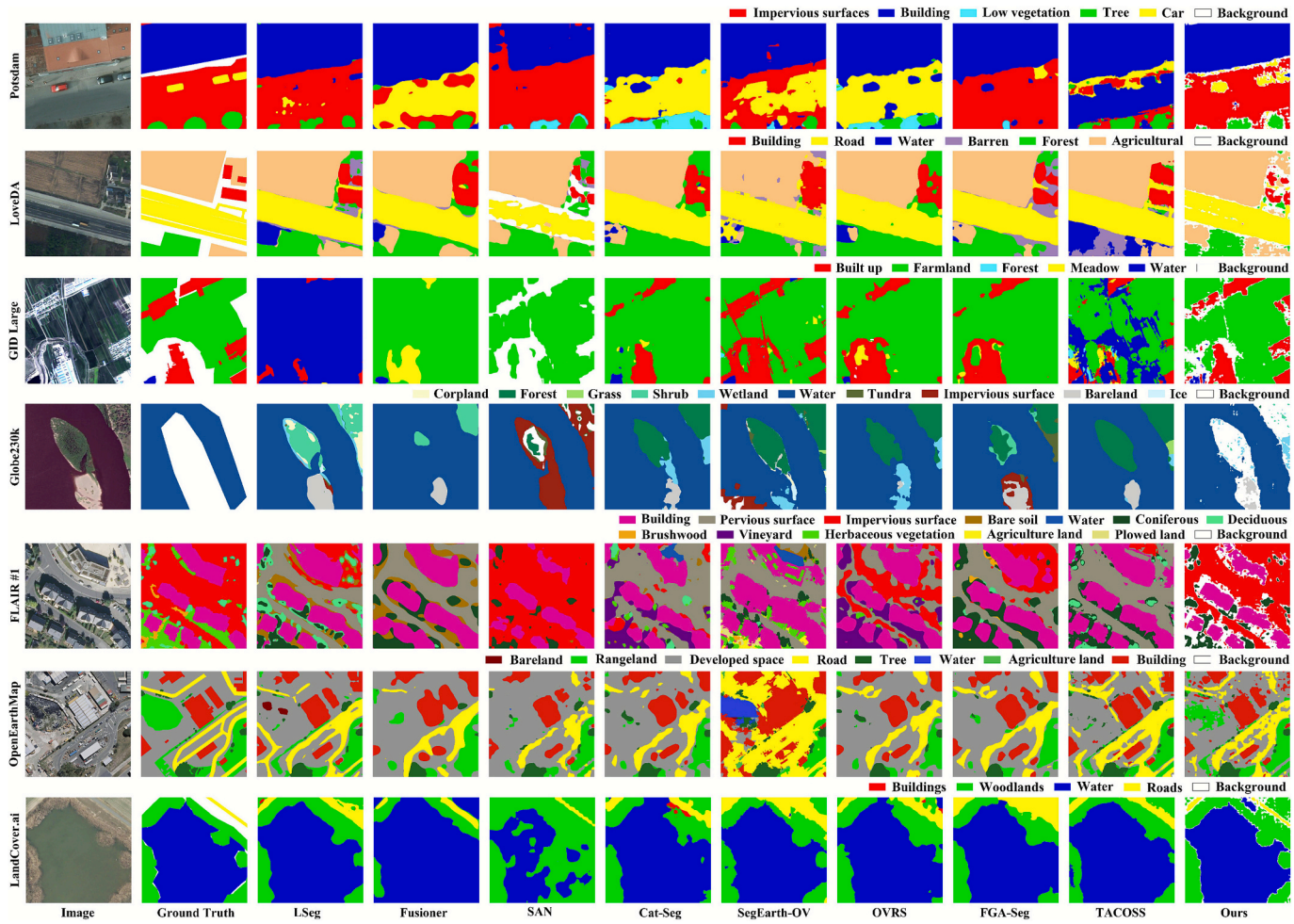

Fig. 8. Qualitative comparison of segmentation results across datasets.

Table 3
Results of ablation studies on different modules. The best and second-best results are highlighted in **bold** and underlined, respectively.

KGCE	CAGD GR	CAGD CAA	BT	*mF*1/*mIoU*(%) Potsdam	LoveDA	GID Large	Globe230k	FLAIR #1	OpenEarthMap	LandCover.ai	*meanmF*1(%)	*meanmIoU*(%)
✓				36.88/26.41	41.54/28.45	43.83/31.52	21.18/13.66	26.12/18.28	68.50/54.23	34.37/22.58	38.92	27.88
✓	✓			48.51/37.23	43.29/30.21	39.07/28.11	26.60/18.25	29.86/21.66	74.00/60.67	45.11/31.97	43.78	32.59
✓	✓	✓		52.37/**40.40**	46.13/32.86	47.81/35.25	32.65/22.77	**32.61**/**23.51**	77.33/64.59	50.77/37.97	48.52	36.77
✓	✓	✓	✓	**53.74**/40.03	52.34/37.74	**62.51**/**46.93**	**34.08**/**23.31**	26.30/19.61	77.33/64.59	55.27/**42.03**	**51.65**	**39.18**
	✓	✓	✓	41.67/30.65	**56.34**/**41.39**	56.34/41.57	21.70/14.68	19.42/13.20	**77.65**/**64.97**	**55.44**/40.47	46.93	35.28

51.65 % *meanmF*1 and 39.18 % *meanmIoU*, outperforming the second-best approach (FGA-Seg) by 8.06 % *meanmF*1 and 6.52 % *meanmIoU*, respectively. While SegEarth-OV, trained without fine-tuning, shows the lowest performance, this is mainly because other models were trained on the OpenEarthMap train set and thus perform particularly well on its test set. After removing this dataset, SegEarth-OV outperforms classical OVSS methods on average across other datasets. Although the two-stage method (TACOSS) is designed for remote sensing, performs the worst on datasets with unseen classes and large domain gaps. This is due to its contrastive learning mechanism on the training dataset, which causes it to focus excessively on specific scenes and lose generalization from the pre-trained model. In contrast, our method consistently achieves top or second-best performance across all datasets, especially on GID Large and Potsdam datasets, where it significantly outperforms other methods, demonstrating its capacity to learn generalizable and discriminative features under various land cover types.

Fig. 7 illustrates the class-wise IoU performance of different models across all datasets. It can be seen that the distribution of IoU values predicted by each model for each class on the same dataset has similarity. Our model achieves the best accuracy in almost all land cover classes, especially in distinguishing between foreground and background. Notably, our model improves or maintains foreground class accuracy while significantly boosting background class accuracy. For instance, in both seen classes (e.g., building, road, water) and unseen

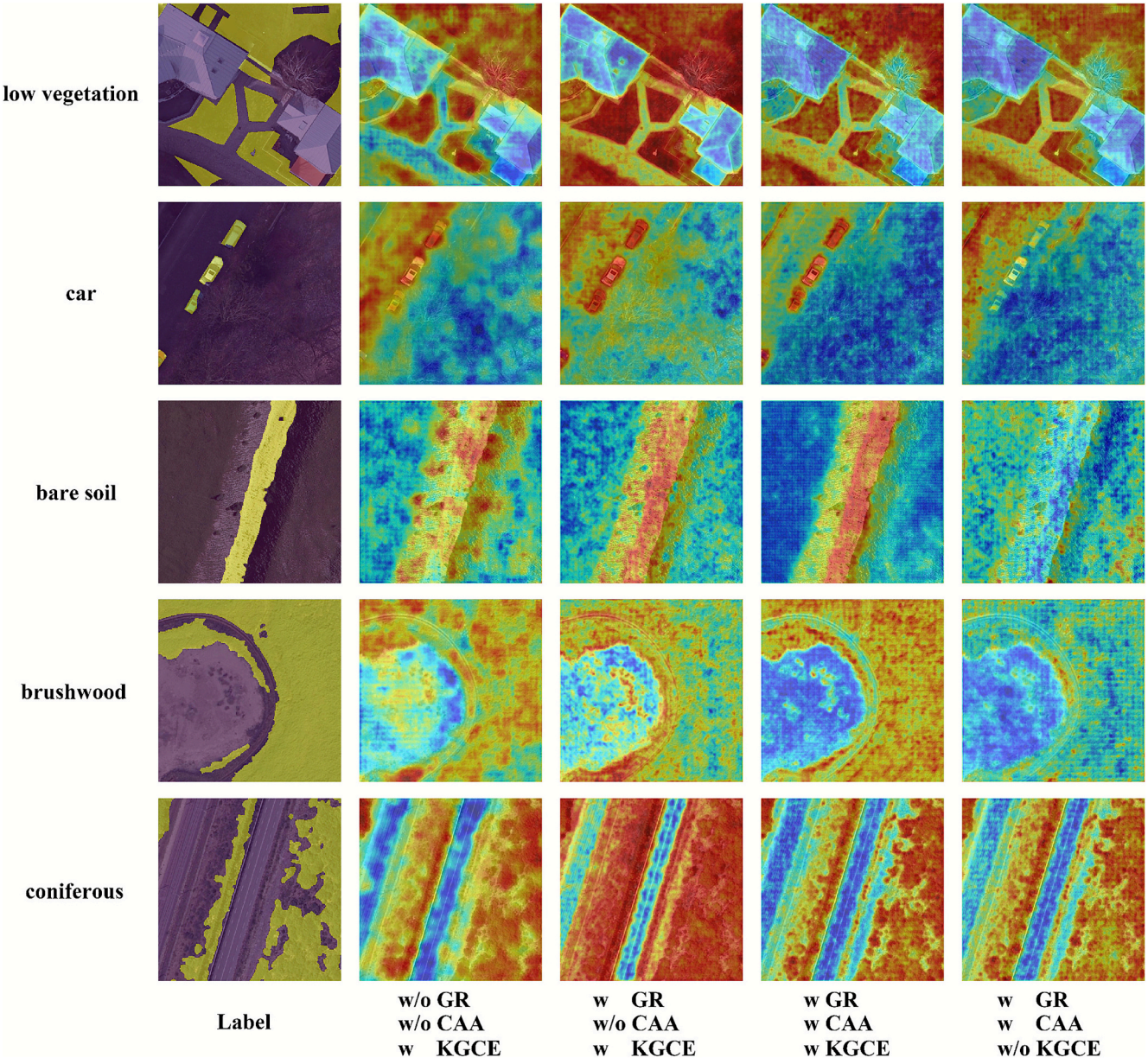

Fig. 9. Feature heatmaps of each module in the proposed method under open-vocabulary settings.

Table 4
Results of different attention methods. The best and second-best results are highlighted in **bold** and underlined, respectively.

Setting	*meanmF*1 (%)	*meanmIoU* (%)
Cross-attention	42.05	31.27
Ours (text(Q), image(KV))	<u>45.30</u>	<u>34.33</u>
Ours (image(Q), text(KV))	**48.52**	**36.77**

classes (e.g., car in Potsdam dataset, forest and meadow in GID Large dataset, impervious in Globe230k dataset, bare soil in FLAIR #1 dataset), our method demonstrates significant improvement. Even for rare classes like wetland, tundra, and shrub, our model maintains relatively stable performance. In contrast, traditional methods perform poorly in most classes, with significant fluctuations in accuracy.

Fig. 8 shows qualitative visualization comparisons. Our method effectively identifies small-object classes (e.g., car), which are often misclassified as background or adjacent classes in other methods. Meanwhile, our method can also distinguish well on some fine-grained classes (e.g., forest, agricultural). For extremely rare classes (e.g., tundra, ice), which other methods tend to confuse with the background, our approach can distinguish them to some extent. However, since the foreground-background separation in our framework relies on a fixed background threshold, pixels with low confidence near class boundaries are often suppressed. This leads to suboptimal smoothness and closure of the predicted boundaries.

Although the proposed method achieves overall state-of-the-art performance, it attains second-best results on the OpenEarthMap, FLAIR#1, and LandCover.ai datasets. This can be attributed to differences in model architectures and data distribution characteristics. Specifically, the LSeg model performs text-image feature alignment in the late decoding stage and heavily depends on the semantic space of the training set, which enables strong performance on OpenEarthMap, but results in limited cross-domain generalization. The FLAIR#1 dataset contains numerous long-tailed classes with low prediction confidence. Under a fixed background threshold, low-confidence predictions are easily classified as background, thereby reducing the *mIoU*. As shown in Table 3, when the background threshold is not applied, our method

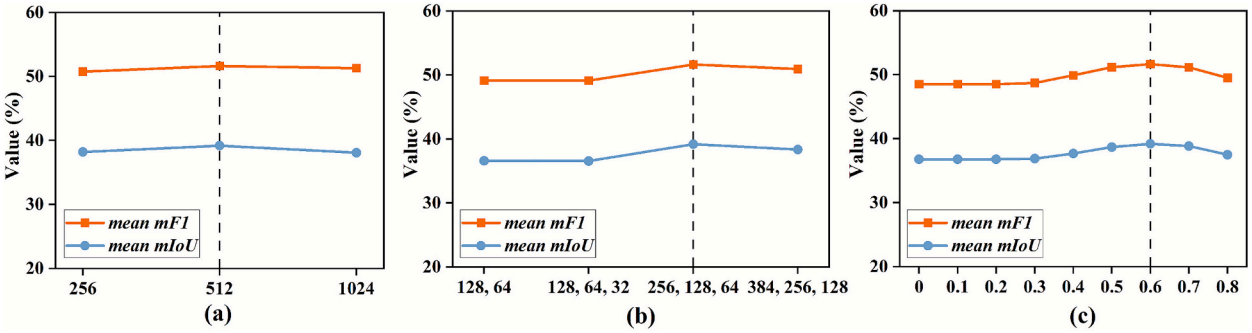

Fig. 10. Performance variations under different hyperparameter settings. (a) Key feature dimension d_k, (b) multi-level image feature channel d_l, (c) background threshold τ.

Table 5
Impact of different relation types r in KGCE on model performance. The best and second-best results are highlighted in **bold** and underlined, respectively.

r	*mF*1/*mIoU*(%)							*meanmF*1(%)	*meanmIoU*(%)
	Potsdam	LoveDA	GID Large	Globe230k	FLAIR #1	OpenEarthMap	LandCover.ai		
synonym	36.39/26.58	52.12/37.36	51.51/38.47	23.05/14.39	23.09/16.40	77.23/64.56	53.26/39.33	45.24	33.87
contain	46.90/34.19	**53.55/38.60**	60.22/44.24	27.57/18.51	25.47/19.47	**77.55/64.86**	55.13/41.25	49.48	37.30
synonym and contain	**53.74/40.03**	52.34/37.74	**62.51/46.93**	**34.08/23.31**	**26.30/19.61**	77.33/64.59	**55.27/42.03**	**51.65**	**39.18**

Table 6
Model complexity and efficiency analysis of different methods. The best and second-best results are highlighted in **bold** and underlined, respectively.

Method	Parameters (M)	GPU memory usage (MB)	First inference time (ms)	Stable inference time (ms)
LSeg	203.752	2530	993.176	37.156
Fusioner	161.334	2540	1023.345	41.277
SAN	436.027	3694	928.199	52.228
Cat-Seg	154.289	2332	930.726	40.325
SegEarth-OV	149.914	**1900**	**430.117**	187.483
OVRS	154.312	2416	950.941	76.095
FGA-Seg	163.021	2458	992.692	45.189
TACOSS	**67.865**	2632	659.731	**31.101**
Ours (w/o KGCE)	159.648	2348	937.351	32.665
Ours (w KGCE)	159.648	2674	2624.143	33.890

achieves an *mF*1 of 32.61 % and an *mIoU* of 23.51 %, significantly surpassing other approaches. This observation suggests that a fixed threshold may introduce bias when the class distribution is highly imbalanced. On the LandCover.ai dataset, the SAN method slightly outperforms our approach, primarily in the background class (see Fig. 7). This is because LandCover.ai dataset contains relatively simple foregrounds with fewer classes. The side-adapter design of SAN is less effective in representing complex foregrounds, causing many pixels to be classified as background, which results in a background bias that coincidentally improves its overall performance. In contrast, our method exhibits notable advantages on datasets with complex foreground structures, where the knowledge graph-enhanced class representations enable more effective differentiation between complex foregrounds and background regions.

5.2. Ablation study

We conducted comprehensive ablation experiments to evaluate the specific contributions of each key module in the proposed KG-OVRSeg to the segmentation performance. We used a baseline model that includes only the KGCE module and excludes the CAGD and BT modules. We then incrementally added the GR, CAA, and BT modules to analyze their independent and collaborative contributions. Furthermore, we replaced KGCE with the standard CLIP text encoder to evaluate the significance of knowledge graph-enhanced semantic embeddings.

As shown in Table 3, all modules significantly improve segmentation performance. The baseline model yields 38.92 % *meanmF*1 and 27.88 % *meanmIoU*. Adding the GR module improves *meanmF*1 by 4.86 % and *meanmIoU* by 4.71 %, indicating that hierarchical guidance during decoding is crucial for fine-grained semantic refinement. Adding the CAA module further boosts performance by 4.74 % and 4.18 %, showing its effectiveness in enhancing class-spatial feature alignment via class-aware attention. Applying the BT module during inference contributes an additional 3.13 % *meanmF*1 and 2.41 % *meanmIoU*, indicating that BT module can further enhance the discrimination between background and foreground in unknown datasets. Replacing the KGCE module with the standard CLIP text encoder results in a performance decrease of 4.72 % *meanmF*1 and 3.9 % *meanmIoU*, highlighting the crucial role of structured knowledge in alleviating semantic ambiguity and enhancing generalization. In summary, all modules positively impact performance, with the GR module being the most critical, followed by CAA and KGCE, while BT provides practical value in stabilizing prediction boundaries and enhancing generalization.

To further validate these findings, we visualized the heatmaps of intermediate features under different module configurations in Fig. 9. It can be seen that introducing the GR module significantly enhances spatial localization, with clearer and more focused activations (e.g., low vegetation and brushwood). After further adding the CAA module, the

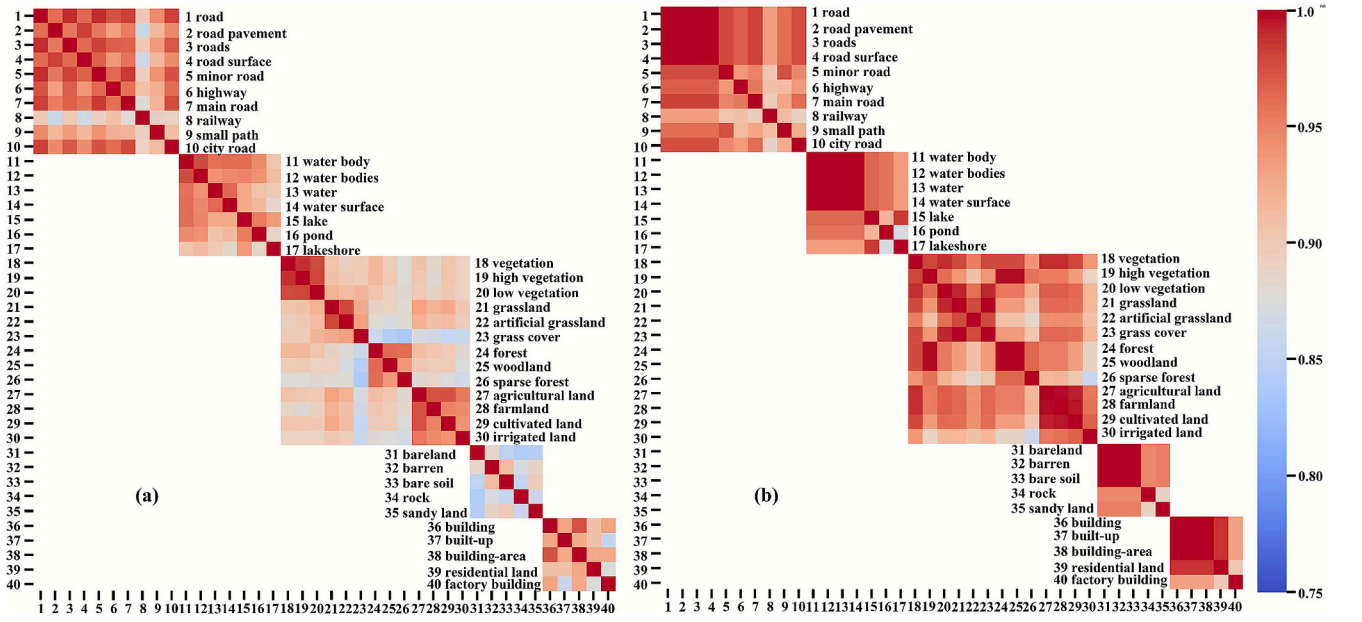

Fig. 11. Cosine similarity heatmaps of some class word embeddings using CLIP ViT/B-16. (a) w/o KGCE, (b) w KGCE.

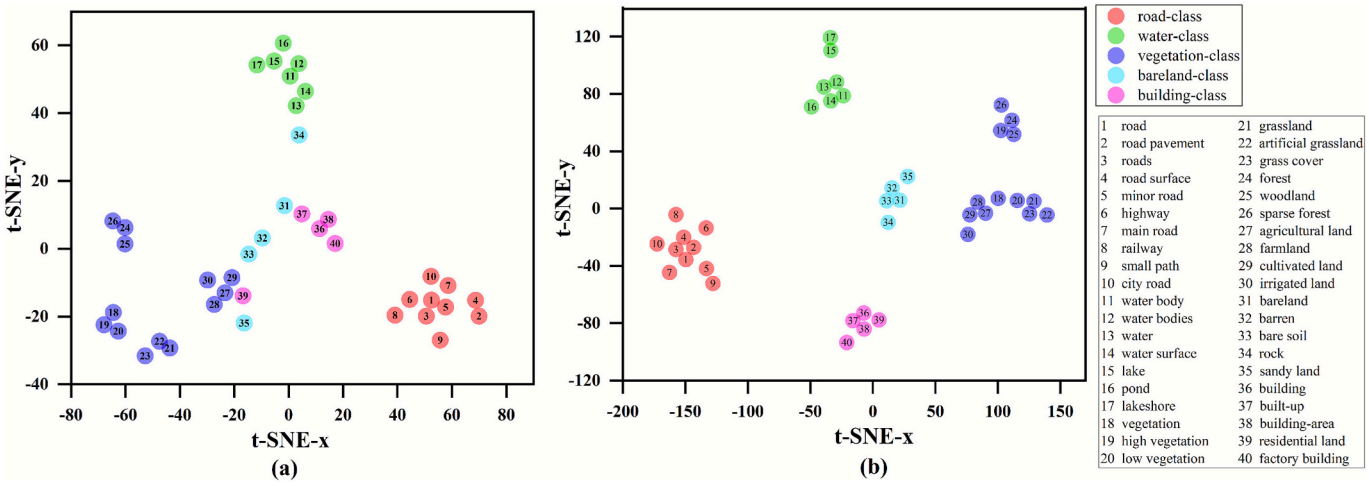

Fig. 12. T-sne visualization of some class word embeddings. The same color indicates that these class words belong to the same class groups. (a) w/o KGCE, (b) w KGCE.

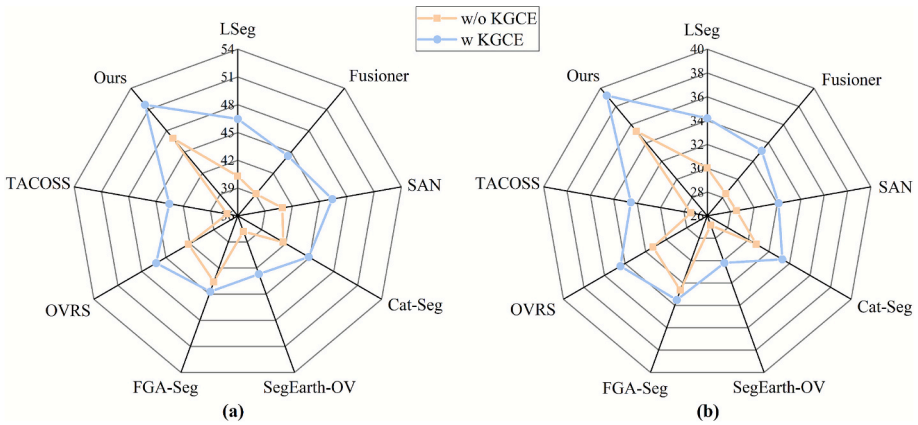

Fig. 13. Quantitative performance comparison of different methods before and after KGCE. (a) *meanmF*1 (%), (b) *meanmIoU* (%).

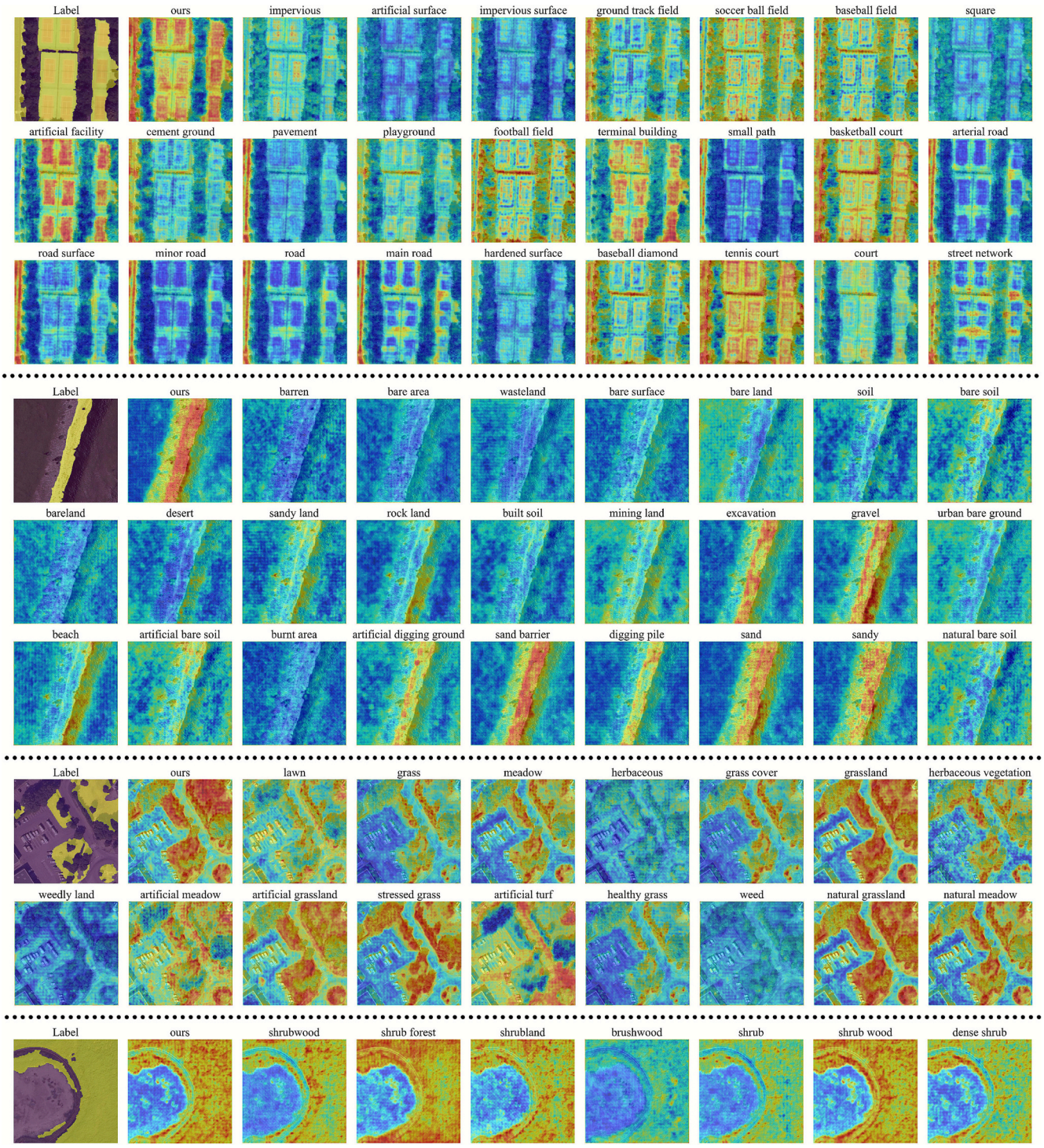

Fig. 14. Response maps between single class words and image features. From top to bottom: impervious surface, bare soil, herbaceous vegetation, brushwood.

semantic discrimination ability of the model was significantly improved. The background interference was significantly suppressed, and the foreground related regions activation was more coherent and accurate. When removing KGCE, we observe degraded or shifted activations, especially for classes such as car, bare soil, and brushwood, illustrating that the KGCE module effectively enhances the semantic perception ability of these classes by introducing structured knowledge, thereby greatly alleviating semantic ambiguity. For the proposed CAA module, we compare its performance with the cross-attention mechanism and with different query-key-value configurations, as shown in Table 4. The proposed CAA with image as Q and text as K, V achieves the best performance, clearly outperforming both the cross-attention and the CAA with text as Q and image as K, V. This demonstrates that using spatial features as query and class embeddings as key and value enables the model to focus more effectively on class-aware spatial regions. In contrast, when text embeddings are used as queries, the attention tends to capture global semantic correlations rather than precise spatial localization.

In summary, the GR module enhances the spatial perception ability, the CAA module strengthens the semantic focusing ability, and the KGCE module solves the problem of semantic uncertainty. The collaboration of the three can effectively bridge vision-language

representations, significantly boosting segmentation performance in remote sensing OVSS tasks.

5.3. Parameter analysis

We further analyzed key parameters that may affect model performance and determined optimal configurations:

(1) Key value feature dimension d_k in CAA Module: This parameter determines how effectively class embeddings attend to visual features. We conducted experiments on d_k with values of 256, 512, and 1024, while keeping other parameters fixed. As shown in Fig. 10(a), the best results are achieved when $d_k = 512$. When increasing or decreasing this value, the model performance will decrease by about 1 %.

(2) Multi-level image feature channels d_l: This parameter controls the richness of hierarchical features. We conducted experiments on d_l with values of (128, 64), (128, 64, 32), (256, 128, 64), and (384, 256, 128), while keeping other parameters fixed. As shown in Fig. 10(b), using three intermediate guided image features with $d_l = (256, 128, 64)$ yields the best results. Configurations like (128, 64) or (128, 64, 32) result in similar but slightly lower performance (~2.5 % difference). Increasing d_l also increases training time and resource requirements.

(3) Background threshold τ: This controls the discrimination between background and foreground during inference. As illustrated in Fig. 10(c), performance remains stable when τ is between 0.1 and 0.3. When τ exceeds 0.3, the metrics begin to rise and peak at $\tau = 0.6$. Beyond this value, performance begins to decline. Therefore, when τ is set to 0.6, the model achieves the best balance between background suppression and foreground preservation.

(4) Predefined relation types r in KGCE: Different relation types affect the subgraph extracted for knowledge enhancement. Table 5 presents the experimental results of train and inference using different r. When only using synonym for knowledge graph enhancement, all metrics of the model are the worst, with *mean mF*1 and *meanmIoU* being only 45.24 % and 33.87 %, respectively. Using contain relations improves performance by 4.24 % and 3.43 %, respectively. Combining both yields the best performance, with an additional gain of 2.17 % and 1.88 %. This demonstrates that contain relations contribute more significantly, and richer relation types lead to more accurate semantic embeddings.

5.4. Complexity and efficiency analysis

Table 6 summarizes the model parameters, GPU memory usage, and inference time per image for the proposed method and other methods. All inference experiments were conducted on a single NVIDIA RTX 4090D GPU. Four images of size 512 × 512 were selected from the OpenEarthMap dataset for the inference, and both the inference images and segmentation classes were kept fixed across methods. In our method, class encoding is executed only during the first inference. Once the encoding is completed, it is cached and directly reused in subsequent inferences. Therefore, the inference time for the first image is reported as the First Inference Time, while the average inference time for the remaining images is reported as the Stable Inference Time.

Our method maintains moderate model parameters and GPU memory usage, comparable to the other methods. Although the first inference takes longer, the average inference time significantly decreases afterward, reaching a relatively low level. This improvement stems from the use of a class encoding cache pool, which encodes the same list of class tokens only once. To further evaluate the impact of the KGCE on model complexity and efficiency, we also analyzed the corresponding metrics without using KGCE. The results indicate that introducing KGCE does not increase the number of model parameters. Although GPU memory usage shows a slight increase, it remains within an acceptable range. The first inference time increases by approximately 2.8 times, which is mainly due to unlike traditional approaches that encode a single class token, KGCE performs encoding over subgraph nodes, involving a larger set of class-related tokens. Nevertheless, the low stable inference time effectively mitigates this overhead.

Overall, the proposed KGCE module introduces only a moderate and controllable computational cost, substantially enhancing the model's semantic reasoning capability and stability. Compared with existing methods, our approach achieves a more favorable trade-off among accuracy and efficiency.

6. Discussion

6.1. Analysis of the effectiveness of knowledge graph encoder

To comprehensively validate the effectiveness of the proposed KGCE module, we conducted analyzes the effect of KGCE from the aspects of semantic similarity and feature visualization between class embeddings. At the same time, the performance before and after introducing KGCE in different methods is compared to demonstrate the scalability and necessity of KGCE.

(1) **Semantic similarity analysis among class embeddings:** We computed cosine similarity between word embeddings of some class pairs and visualized the results as a heatmap, as shown in Fig. 11. After introducing KGCE, we observe significantly increased similarity among subclasses within the same class groups. Subclasses such as barren, bareland, and bare soil, or grassland and grass cover, exhibit near-perfect similarity, indicating that KGCE effectively alleviates inconsistencies caused by variations in class naming. This demonstrates KGCE's strong capability in aligning semantically related classes within the same conceptual group.

(2) **Visualization of category embedding feature distribution:** Fig. 12 presents the t-SNE visualization of semantic embeddings under different settings. Without using KGCE, some semantically similar classes (e.g., residential land and building-area, rock and bareland) are distant in the embedding space, and subclasses under class groups like bareland, building, and vegetation are sparsely distributed with blurred boundaries. In contrast, with KGCE, intra-class clusters become more compact and inter-class separability improves significantly. This shows that KGCE effectively enhances the discriminability of class semantic representations.

(3) **Performance comparison of different OVSS methods using KGCE:** Fig. 13 shows radar plots comparing the *meanmF*1 and *meanmIoU* of different OVSS models before and after integrating KGCE. The results show consistent performance improvements across all methods, with *meanmF*1 and *meanmIoU* increasing by an average of 4.6 % and 3.5 %, respectively. Notably, TACOSS has the greatest improvement, with *meanmF*1 and *meanmIoU* increasing by 6.36 % and 5.15 %, respectively. Furthermore, our method consistently achieves the highest performance regardless of whether KGCE was used, confirming the robustness of our framework again. These results strongly support the general applicability and effectiveness of KGCE in enhancing semantic modeling across different OVSS pipelines.

6.2. Analysis of the response relationship between class words and image features

We visualized response heatmaps for various class words and image features to further investigate the influence of class word embeddings on the activation of image features. As shown in Fig. 14, we selected four representative classes: impervious surface, bare soil, herbaceous vegetation, and brushwood. During inference, our model inputs multiple subclass words from the KGCE subgraph (used for representation enhancement) to generate corresponding response maps. These results (generated without using KGCE, using only single-class words) are compared with the full KGCE-enhanced result (Ours).

When using a single class word for inference, the responses tend to be fragmented and less focused, with blurred boundaries. In contrast, KGCE leads to more concentrated and accurate activation regions. Moreover, for some class words annotated in the dataset (e.g., impervious surface, bare soil, herbaceous vegetation, and bushwood), the single-word embeddings result in poor or even no activation, indicating semantic ambiguity.

For instance, impervious surface contain subclasses like road, court, and roof, which may not be effectively captured by a single class word. KGCE includes these subclasses in its enriched embeddings, significantly improving performance. Visualizations of subclass responses reveal that road-related words strongly activate road regions, while artificial facility and tennis court respond well to courts. Similar trends are observed in the bare soil class, where words like sand barrier and sand show superior response over the labeled class word and their synonyms.

In the results of herbaceous vegetation and brushwood, although there is some response to the labeled class word, synonym words like grassland and shrub wood produce more accurate and stronger activations. This variation highlights semantic inconsistency even among synonyms, reflecting the limitations of relying on single-word embeddings.

In summary, the above findings demonstrate that different class words exhibit varying levels of responsiveness to the same image region. And the single-word embeddings are often insufficient due to synonym diversity and semantic ambiguity. KGCE, by integrating subclasses and synonyms into a unified embedding, provides richer and more accurate semantic guidance. This results in clearer, more focused region activations and significantly enhances OVSS performance.

6.3. *Limitations and future work*

Although the proposed KG-OVRSeg framework demonstrates strong performance and generalization in OVSS for remote sensing imagery, there are still some limitations that deserve further research. First, the performance of the KGCE depends on the completeness and accuracy of LCCKG, which serves as its structured semantic foundation. Despite covering a wide range of land-cover classes, LCCKG may not include all possible classes, particularly region-specific or emerging classes. When encountering out-of-graph classes, the model's performance tends to revert to that of the baseline model without knowledge enhancement. Future work could explore dynamic KG expansion, using LLMs for automatic concept extraction and relation mining to continuously maintain comprehensive semantic coverage and enhance class representations as new classes emerge. Second, our framework employs a global fixed background threshold to further distinguish foreground and background pixels during inference. This design introduces two challenges: (1) suppression of low-confidence pixels near object boundaries, leading to unsmooth or incomplete edges (See Fig. 8), and (2) suboptimal performance on long-tailed datasets (e.g., FLAIR#1), where rare classes are easily misclassified as background (See Table 3 and Fig. 7). Future studies could explore adaptive or spatially varying threshold strategies to generate sharper boundaries and better balance across classes. Finally, this study primarily focuses on optical remote sensing imagery. Extending our knowledge-enhanced segmentation paradigm to multimodal data (e.g., SAR, LiDAR, hyperspectral) and broader visual understanding tasks represents a promising direction for future research.

7. Conclusion

In this study, we proposed KG-OVRSeg, a specifically designed framework for remote sensing OVSS, which effectively addresses semantic ambiguity and uncertainty problems such as one-to-many, many-to-one, and weak embedding proximity. By introducing a KGCE, the model incorporates subclass and synonym information into the class representations, thereby enhancing semantic discriminability and robustness. Additionally, the CAGD progressively aligns visual features with class-related regions using CAA and GR strategies, effectively suppressing background noise and sharpening fine-grained class boundaries. For complex background classes, the BT strategy is applied to distinguish foreground and background during the inference. Extensive evaluations on seven different datasets demonstrate that KG-OVRSeg outperforms state-of-the-art OVSS methods. It not only improving segmentation accuracy on seen classes, but also significantly enhancing recognition of fine-grained and rare unseen classes. Furthermore, the KGCE module greatly enhances the semantic coherence among class embeddings and the accuracy of its spatial responsiveness to class features. The generality and modularity of KGCE make it easily integrable into existing remote sensing OVSS frameworks. This work lays a foundation for robust and adaptive interpretation of open-world scenarios in remote sensing. Future work will focus on optimizing the proposed framework through dynamic KG expansion and adaptive thresholding strategies, as well as exploring its effectiveness on multimodal remote sensing data.

CRediT authorship contribution statement

Wubiao Huang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Huchen Li:** Writing – review & editing, Validation, Software, Investigation, Formal analysis, Data curation. **Shuai Zhang:** Writing – review & editing, Visualization, Software, Investigation, Data curation. **Fei Deng:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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